

Simulation of demand growth scenarios in the Colombian electricity market: An integration of system dynamics and dynamic systems

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HIGHLIGHTS

- A simulation model was constructed based on SD and DS approaches.
- Seasonality and growth demand uncertainty on the Colombian power market are studied.
- The DS/SD integration provides further understanding and a new and broad analysis methodology.
- Colombia needs new capacity before 2019, otherwise it might undergo rationing events during 2020/2021.

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ABSTRACT

Modeling and simulation of electricity markets have increasingly involved the use of a system dynamics (SD) approach. Accordingly, the resulting dynamic hypothesis and the stock-flow structures are represented and simulated using softwares such as Stella, Powersim, iThink, or Vensim. However, SD models can be exploited even more, of which the investigation of signals in the time domain or the sensitivity analysis is just a small part of the study. Since SD models are mathematical objects, they deserve an analytical or numerical study using tools provided by the dynamic systems (DS) methodology. Therefore, this paper not only studies the dynamic hypothesis or the stock-flow structure of an electricity market model in the classical form, but also uses its inner mathematical object to provide a deeper insight into the system. Using MATLAB/Simulink®, the system is evaluated from a different approach not yet reported in the literature. The combination of the SD and DS methodologies can open the door to a new and alternative method of analysis for electricity market models and even for any SD model. In fact, this paper demonstrates that with this methodologies combination, more detailed analysis strategies and novel insights of the SD models can be developed, which can be easily exploited by policy makers to suggest improvements in regulations or market structures. Moreover, considering that the energy market is evolving, and a series of macro and microstructural changes are impacting demand, we consider as an example a simplified version of the Colombian electricity market to report a detailed description of its dynamics under a broad range of growth rate of demand (GRD) scenarios. Our study, inspired by the bifurcation and control theory of DS, primarily shows that Colombia is in dire need of a new capacity before 2020 to avoid rationing events expected to occur in the upcoming years.

1. Introduction

Simulation plays an important role in academia and industry. It is well known that experimentation with real systems is not possible in most cases given the undesired consequences or high costs that can be incurred. In particular, in the case of electricity markets, simulations are the basis for policy design and regulation improvements. Simulations allow the suppliers to invest in the best energy mix to stimulate the appropriate market price and to meet adequately the energy

demand, among others.

In particular, the simulation of electricity markets has been increasingly addressed with the system dynamics (SD) approach. Currently, in the literature, countless papers that investigate the different schemes of electricity markets under the SD approach can be found. For instance, energy efficiency [1,2], renewable technologies diffusion [3–5], capacity adequacy [6], policy planning [4,7], demand response [8], among others, are topics of great interest for the SD community. In fact, the SD methodology has been applied successfully

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and many important works have been developed accordingly [9]. The most remarkable advantage provided by the SD technique is the ability to efficiently capture the complex structure of real systems under a holistic overview. Indeed, modelers who are not familiar with mathematical models can find it easy to represent their problems using the SD approach. In this sense, modeling complex systems such as electricity markets have evolved from a simple stock and flow diagrams to large and hybrid models, involving engineering optimization, genetic algorithms, decision-tree approaches, and agent-based modeling [9]. This tendency to mix SD models with other strategies enhances the overall analysis, provides deeper insights, and covers more variables or scenarios. In essence, SD has shown to be compatible with other modeling techniques.

In fact, it is clear within the SD community that Forrester created SD inspired by the control theory and that an SD model always involves an indirect treatment with ordinary differential equations. In this sense, the formulation of a model in the SD world corresponds to a dynamic system or a set of ordinary differential equations; therefore, the compatibility of SD with other modeling techniques lies on the fact that an SD model is a mathematical object, and the dynamic systems (DS) community is aware of the immense mixing possibilities available to address an DS model. This is significant because the use of hybrid modeling techniques always leads the scientific community to further exploration and understanding of the nonlinear behaviors and feedback relations of the complex systems.

Nevertheless, literature is lacking not only in the electricity sector modeling, but also in other disciplines that integrate the SD modeling technique with different tools provided by the DS methodology, that extend the reduced routes of analysis to a broader spectrum of possibilities, only limited by the modeler knowledge. In this sense, our paper seeks to demonstrate the importance of integrating DS tools in the classic SD modeling cycle aimed at providing deeper understanding and insights into the SD models, as well as showing that this combined methodology can be implemented by using much simpler steps, i.e., without resorting to complex mathematical processes. In fact, the integration of SD and DS is not a new trend. Javier Aracil pioneered the stability concept applied to SD models from the DS perspective at around 1980 [10]. Subsequently, in [11] Aracil officially reported the importance of studying the qualitative behavior of SD models under their mathematical properties, to provide a more solid foundation of the analysis; however, his work has not been given much attention by the SD community since then. However, a few years ago, some authors have raised Aracil's investigations again. In [12–14], the dynamics of small electricity market models were described using the DS perspective. In particular, the set of dynamic equations were studied analytically and programmed in MATLAB® to investigate the bifurcation regimes in electricity markets for the first time. However, their proposed models were a small version of the real one and many assumptions were considered. Unfortunately, by considering the small-version models and neglecting several important properties of the electricity markets, the results are not accurate enough and the corresponding conclusions may change. In addition, although working directly with the system equations is possible and feasible in some cases (the system equations can be programmed in any software package to obtain a solution under the numerical methods; subsequently, anything can be developed to analyze the system behavior), most of the models, and even most of the electricity market models have a high level of complexity, involving several feedback relations, state variables, and delays. Therefore, the models are almost impossible or cumbersome to study analytically or even programmed in any software package. Hence, we believe this particular methodology is not awakening the interest of the SD community, who are not familiar with the engineering design cycle since it does not support the way SD modelers think. As a result, in this work we propose the combination of the SD and DS perspectives in a simpler way such that SD modelers can be more comfortable working with DS tools. Thus, we believe that our study in this paper will encourage the SD

community, especially those interested in gaining more knowledge from the SD models, to further their studies to more than just a time series or a limited sensitivity analysis, and to open the door towards an infinite spectrum of possibilities for analyzing SD models.

To pursue our objective, we propose to create, in MATLAB/Simulink®, block diagrams analogous to the classic stock–flow structures that facilitate the deeper insights or analysis of the SD models. In particular, based on the bifurcation theory and the describing functions (DF) commonly used in control theory, we develop an advanced sensitivity analysis, a tool for determining the time of occurrence of the confidence limits and a scheme for describing in great detail the rationing events of our system. As an example, we investigate a simplified version of the Colombian electricity market; in particular, two important issues are analyzed: (i). How the variability of hydrogeneration affects its performance and (ii). What should Colombia be aware of under different growth rate of demand (GRD) scenarios. It is noteworthy that our developed methodology can be used for the electricity market models or any model of any disciplines regardless of its level of abstraction.

Although Simulink was already used in [15] to show the compatibility of the stock–flow structures with the block diagrams modeling, the authors did not implement the complete model (as we propose) in Simulink; they only linked MATLAB and Vensim to take advantage of MATLABs capability to manage the algebraic constraints. More recently, other authors have also developed algorithms in MATLAB for computing certain complex functions and then sent them to Vensim for subsequent processing [16,17]. In other words, the authors implemented some functions in MATLAB for computing, with more precision and simplicity, complex algebraic expressions of the model. However, from our perspective, linking MATLAB with SD software packages is cumbersome and also restricts the full potential of the MATLAB tools.

Furthermore, it is noteworthy that the SD/DS simulation through function blocks has been already introduced in other disciplines not related to the electricity markets. For instance, using Simulink for SD modeling was briefly mentioned in [18] for comparing the three major paradigms in simulation modeling: SD, discrete event, and agent-based modeling. Nevertheless, it was first reported in [19] that an SD model was converted into DS function blocks for combining genetic algorithms parameter search, fuzzy logic expert input, and SD modeling, to a very general market growth model. Similarly, in [20] the potential of block diagrams modeling was also used for an SD supply chain model with a capability limit. Finally, in [21], the Simulink block diagram modeling of an SD manufacturing supply chain model with uncertain demand was utilized once again. Even though all these works show that the SD/DS simulation through function blocks results in better strategies for decision making, since this methodology provides much more freedom in customizing and implementing global analysis techniques, they do not suggest the methodology for other disciplines or consider it as a new research trend in the classical SD modeling cycle. Indeed, in the referenced papers, bifurcation or control theory concepts were not considered for their research.

On the other hand, the variability in the generation of renewable sources has been addressed in several countries, especially for solar and wind energies. However, intermittency has been used more frequently when referring to the variability in these sources of generation. In fact, intermittency has been primarily addressed in Germany [22], the Alpine region [23], Spain [24], Canada [25], USA [26], and China [27]. In summary, these works state that intermittency highly affects the electricity tariff, demand and supply, investments, reliability, and dispatchability of electricity markets. However, the variability in hydrogeneration has only been addressed in Africa [28], USA [29] and China [30]; in particular, climatic variability issues in Colombia has been studied but only for explaining hydroclimatic anomalies and their impacts on biodiversity, ecosystems, and global environmental change [31]. Evidently, the variability in hydrogeneration and its impacts on

the Colombian electricity market performance have not been investigated yet.

Furthermore, not many studies can be found in the literature that involves *GRD* scenarios. For instance, a sensitivity analysis on demand was performed in [32]. The results showed that the Netherlands was very robust and therefore, rationing events were not expected in the short run. However, the authors applied a simple sensitivity analysis for only two scenarios using an unknown simulation model. More recently, in [33], four demand scenarios were considered in a sensitivity analysis, again performed on the Netherlands power market; the proposed model, based on a linear optimization problem, concluded that the installed capacity would not be affected and the market conditions would, in fact, become more favorable. Once again, the sensitivity analysis was very simple and only a few scenarios were investigated. In fact, we found that the demand response against different market conditions is currently an investigation trend [34], but this topic is out of the scope of the present work. Clearly, there is a lack of literature that studies a broad range of *GRD* scenarios and especially, that which combines SD and DS and aimed at developing more advanced tools of analysis to provide deeper insights into the SD models. In addition, the investigations on these important topics involving Colombia are not found. In this sense, the present paper also seeks to investigate how changes in the *GRD* can affect the performance of the Colombian electricity market. However, we believe that some of the lessons from the Colombian case might be applicable to other situations or countries.

2. Overview of the Colombian electricity market

This section describes the main characteristics of the Colombian electricity market that have been considered for developing our simplified model. First, it is important to highlight that Colombia is gifted with natural resources, reflected in an electricity mix mostly dominated by hydro power generation (see Fig. 1) with approximately 70% (≈ 11611.1 MW) of its total installed capacity, followed by fossil fuel plants, and non-conventional renewable sources that represent approximately 29% (≈ 4833 MW) and 1% (≈ 151 MW), respectively. Hence, we can conclude that Colombia is primarily driven by hydro and thermal power plants. In this sense, we can assume in our model that the electricity demand (*ed*) of the Colombian power market is basically met with hydro and thermal generations.

Since the liberalization of the energy market in 1990 to promote private investments, Colombia has achieved a clear legal framework, fair competition conditions, stability for investors, and improvements to the security of supply [36]. Inspired by the British model, the Colombian electricity sector has been reformed throughout the years and is now more efficient and reliable owing to the well-developed laws that promote a good market competition. As a result, a very clean energy mix has been growing and now relies on 70% hydrogeneration, as mentioned above.

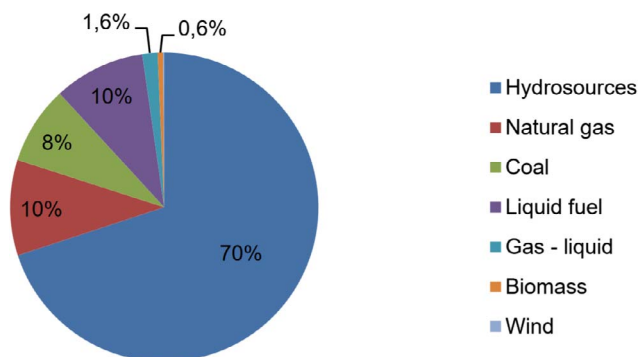


Fig. 1. Colombia's installed capacity at year end 2016.
Source: Ministry of Mines and Energy [35].

In general terms, the dispatch process to meet the *ed* is based on the lowest price offers. This means that the sources of generation able to produce power at the lowest price possible, are placed at the top level of the electricity matrix to supply first the *ed*. If the *ed* is not fully met, the next source of generation in the electricity matrix is called to supply more electricity. This process continues until the total electricity is supplied. Once the total *ed* is met, the system sets the market price according to the price offer established by the last source of generation used. In other words, the dynamics of the Colombian electricity market is governed by supply and demand forces.

Furthermore, the operation of the hydroplants is highly dependent on the level of their reservoirs, while thermal plants rely on fuel availability. In fact, thermal plants in Colombia are more reliable and are thought of as a support for hydrogeneration, since fuel availability is not a current problem. On the contrary, hydroplants can be strongly affected by weather conditions considering that long periods of droughts is a pressing issue in the country currently.

Although Colombia is proud of the cleanliness of its energy production, a 70% installed capacity of variable generation (hydroplants) can cause serious problems to the electricity sector by virtue of the weather uncertainty. In fact, this variability in hydroplants can be more pronounced when other market variables are altered, for instance, the *GRD*.

3. Simulation overview

The results described in the present paper are obtained by combining the SD and DS perspectives; accordingly, two computer programs are developed. Initially, under the SD approach, the dynamic hypothesis of the Colombian electricity sector that will be addressed in this study is identified. Subsequently, the corresponding stock-flow structure is derived from the causal-loop diagram to perform a more detailed quantitative analysis. Following that, the stock-flow structure is transformed into a Simulink block diagram to analyze the system from a DS perspective.

More precisely, to investigate our system from a deeper insight, it is necessary to follow one of these two steps: (i) Obtain the dynamic equations, perform an analytical study if possible (when the system has a solution), program the equations in any programming language (basically determined by the modeler depending on system requirements, simplicity, interface, user-friendly, performance, accuracy, among others), and finally develop any analysis strategy based on the existing DS tools, or (ii) Create the analogous block diagram of the stock-flow structure obtained from the SD approach in Simulink. In other words, inspired by the control theory, transform the stock-flow structure into a block diagram. Subsequently, develop any analysis strategy in a normal MATLAB script based on the existing DS tools.

In this paper, we decided to follow the second step for simplicity, because the dynamic equations have no analytical solutions (they can only be approximated using numerical methods), and for creating a friendly simulation environment. Accordingly, we create the analogous block diagram in Simulink®, and then implement different analysis tools in a single MATLAB® script that will be discussed below in this paper. The overall model has a simple design that is easy to modify and extend. Indeed, we want to encourage the SD community to combine the SD and DS approaches in their studies such that deeper insights and more sophisticated results can be accomplished.

3.1. Dynamic hypothesis

In this paper, the electricity market model seeks to show the causalities among the different Colombian market variables, the *GRD* scenarios, and the imminent effect of seasonality in the generation process. As shown in Fig. 2, the dynamic hypothesis comprises three balance loops. B_1 represents the dynamic interaction of the demand side variables, while B_2 and B_3 represent the supply side associated to hydro (V)

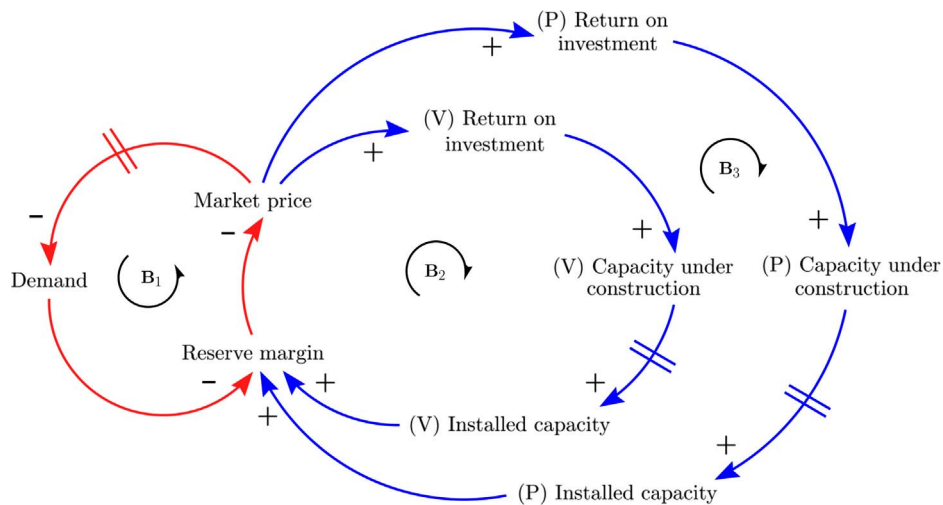


Fig. 2. Causal-loop diagram.

and fossil fuel (P) generations, respectively. As mentioned above, the Colombian electricity mix is dominated by hydrogeneration, which is considered a variable source of generation because it is affected annually by the climate variability in the country; therefore, (V) in Fig. 2 refers to the variable hydrogeneration. Similarly, the second biggest source of generation (thermal power) is considered a permanent source of generation considering that its availability is always constant; as a result, (P) is used to refer to this generation technology.

The balance loop B₁ clearly indicates that the increasing values of market price incentivize the reductions in energy consumption, which in turn affects the reserve margin positively. Similarly, when the electricity market is experiencing a reserve margin shortfall, B₂ explains that consumers have to pay a higher price. However, this results in greater returns on investment for the producers. Clearly, this incentivizes the expansion of both the variable and permanent capacities, since the market price causalities not only affect the balance loop B₂, but also B₃; consequently, the reserve margin is affected positively. It is noteworthy that this increment in the reserve margin balances the subsequent causalities.

3.2. Stock and flow diagram

The second step of our work implies, as the SD approach suggests [37], a stock and flow building process to perform a quantitative analysis. This process allows us to transform the causal-loop diagram into a stock and flow diagram, which describes the system in more detail involving the formulation of the dynamic equations.

First, the supply side of (P) generation (referring to all Colombian fossil fuel power plants) involves the expansion of this sector, comprised of two stock variables: its capacity under construction and installed capacity, as Fig. 3(a) illustrates. The construction of new plants depends on the investment decision of the producers, which is entirely determined by their returns on investment. The higher the electricity tariff, the higher is the investment incentive for new capacity. In Colombia, the highest price is typically reached when thermal plants are used to produce electricity, since the cost of fuel is higher than producing energy with water resources.

Similarly, the supply side of (V) generation (referring to all Colombian hydro power plants) is also associated with the expansion of this sector, as shown in Fig. 3(b). Installing a new capacity is highly dependent on the producers profits. In other words, high electricity tariffs benefit the capacity under construction because the producers desired investment is stimulated, which eventually increase the installed capacity of the hydrogeneration and decrease the risk of blackouts.

Nevertheless, the thermal and hydro power plants, which have been operating for many years will eventually become obsolete, affecting the installed capacity negatively since they need to be removed from the system. In fact, the dynamics of the installed capacity in both technologies of generation are affected similarly by two flows, the retirement of old plants and the retirement of initial ones, as shown in Fig. 3. The flow retirement of the initial plants is used to remove the initial value of the installed capacity, i.e., the current state of the permanent installed capacity, or the current state of the variable installed capacity. The initial installed capacity refers to all hydro and thermal power plants that were previously built and are still in operation. However, determining their lifetime is a very difficult task primarily because most of them became part of the system in different years and others do not have an accurate information of when they started to participate in the Colombian electricity market; as a result, they are smoothly removed from the installed capacity, taking into consideration the average lifetime of a general hydro (or thermal) power plant, using a first order delay [38]. On the contrary, to gain accuracy in the model, a new installed capacity is removed from the system once the power plants become obsolete, using a pipeline delay (infinite order delay) through the flow retirement of old plants [38].

Fig. 4 shows that the power demand plays an important role in the market dynamics. Clearly, the interaction among the generators, who compete for providing energy services at the price set by the market, determines the reserve margin of the electricity system. Furthermore, the market price formation not only depends on the reserve margin, which sets the rationing price when its level reaches a critical value, but also on the last generation technology that is participating in the dispatch process to meet the total demand. In addition, this market price is slightly delayed since the consumers in Colombia perceive the current electricity price with a certain lag. Consequently, and as it is expected in the real system, the market price, together with the elasticity of demand in Colombia [39], cause either a positive or negative effect on the demand behavior.

Finally, the dispatch process is considered in Fig. 5. Assuming a perfect electricity market competition, the firms cannot influence the market price. In fact, the dispatching merit order is determined by the market, which sorts the available generation technologies according to their marginal costs, from the cheapest to the most expensive. In other words, once the supply equals the demand, the market price is set by the most expensive generation technology, which is also running at that time. This price determines the returns on investment, which eventually influence the system capacity expansion, as it was explained in the supply side modeling. Moreover, in the dispatch side some other variables are determined, for example, the utilization factor, which is a

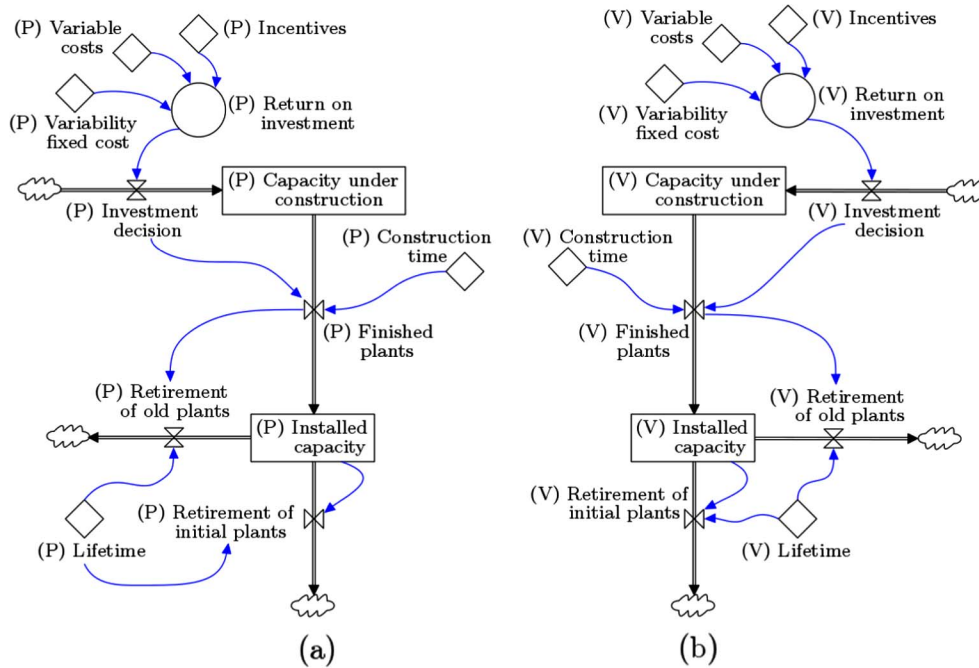


Fig. 3. Supply side from (a) (P) and (b) (V) generation.

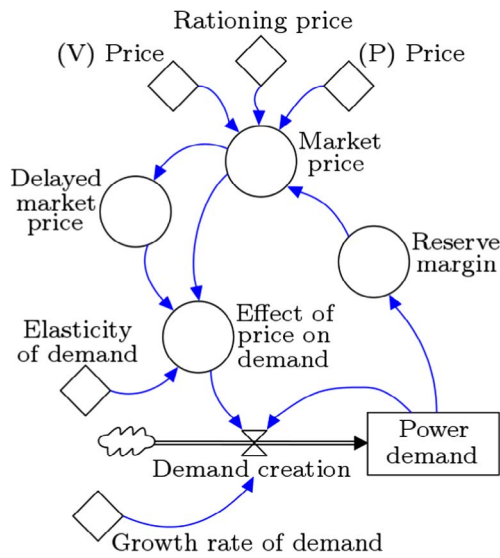


Fig. 4. Demand component.

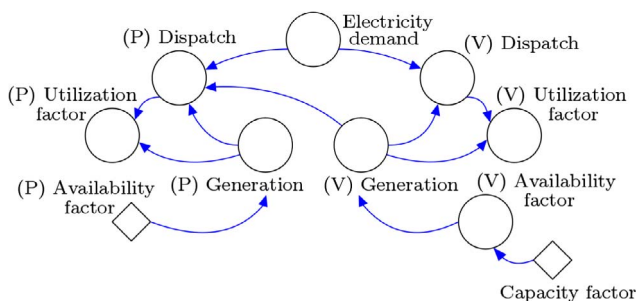


Fig. 5. Electricity dispatch.

percentage measure of the power plants participation in the dispatch process, is used to directly affect the returns on investment together with the market price. In other words, the amount of energy dispatched by each technology per its generation capacity (generation) provides a

rate of usage of this technology (utilization factor), which serves to compute its return on investment. The generation capacity, however, depends on the source of generation, and essentially, on its availability. Thermal power plants for example, are considered permanent sources of generation owing to their almost constant availability that is only restricted by fuel availability; as a result, the availability factor can be kept constant. However, the generation capacity of hydro power plants is highly determined by either the amount of water in the reservoirs or the flow of the rivers, both strongly affected by weather conditions. Consequently, its availability factor cannot be constant and is modeled by taking into account the water contributions of the Colombian rivers, measured as the water flow level entering the system. Although this availability factor can reach values higher than 100%, the capacity factor typically reduces it owing to different technical constraints.

It is noteworthy that all the stock and flow diagrams described in this paper were transformed into the corresponding Simulink block diagrams and together with the model equations, are shown in Appendix A.

3.3. Seasonality

As mentioned above, the electricity mix in Colombia is dominated by hydrogeneration; consequently, the market dynamics is strongly influenced by the Colombian weather conditions, i.e., the seasonality. In essence, these weather conditions are directly transferred to the water contributions of the Colombian rivers, which in turn provide variability to the availability factor of the hydrogeneration. In general, seasonality in Colombia refers to two clearly identifiable times: rainy and dry, which cause variability in the generation capacity. That is, high levels of generation capacity when the weather is cold and rainy (half of the year), and low (sometimes critical) levels of generation capacity when the weather is hot and dry (the other half of the year). By contrast, thermal generation can provide electricity at a nearly constant rate since fuel availability is not affected by external and uncontrollable forces.

It is noteworthy that the resources uncertainty of the renewables is currently a great concern for the energy specialist and energy markets, not only because renewables are characterized as being variable or intermittent, but also because their installed capacity is growing

worldwide, and is expected to continue growing for the next 30 years [40,41], which in turn will increase the resources uncertainty as well as its consequences. In this case, due to the merit order effect together with the fact that Colombia is a hydro dominated country, the resources uncertainty becomes the primary source of complex dynamics exhibited by the electricity market. In fact, the resources uncertainty plays a key role in any electricity market primarily driven by renewable energy sources. Basically, since renewables are not dispatchable the whole market dynamics are affected by this uncontrollable phenomenon, which not only stresses or alters the system, but also affects the customers by making the electricity price more volatile. We will show below how the nearly cyclical behavior of the seasonality directly influences the dynamics of most of the key system variables.

The question arises on how the seasonality dynamics can be incorporated into our model. Since our interest is to study the effects of growth rate of demand variations in the Colombian electricity market, in principle we plan to capture only the primary properties of seasonality that provoke variabilities to hydrogeneration, i.e., capture the nearly cyclical pattern of the weather dynamics. Hence, it is noteworthy that in this case we neglect the ENSO (El Niño–Southern Oscillation) phenomenon.

To model the seasonality, we first obtained from the XMs records [42] the historical average water contributions of the Colombian rivers from January 2000 to December 2016, in terms of energy (kWh). More precisely, this data contains the average water contributions of the corresponding past months from 2000 to 2016. With this information we proceed to recreate the general and almost cyclical behavior of the Colombian weather in an average year. Subsequently, the availability factor of the variable generation is obtained, in percentage, as shown in Fig. 6 through the red line.

For simplicity, and to avoid adding probabilistic terms to our model, we plan to approximate the real data using deterministic functions, such as sine and cosine functions. As shown in Fig. 6, the real availability factor (red signal) can be very well represented by a simulated signal (blue signal) obtained with the following expression:

$$af_v = 1.01 + 0.47\sin(2\pi t - 0.45\pi) \times \cos(2\pi t - 0.45\pi) + 0.25\sin(2\pi t - 0.55\pi) \quad (1)$$

where af_v refers to the availability factor of the variable generation to be used in the following simulations. The parameter values were calibrated using the historical data of the Colombian rivers from 2000 to 2016 to achieve a good representation of the real signal. Moreover, it is noteworthy that this af_v will be used in the whole-time horizon of our model, with which we only expect to characterize the nearly cyclical

behavior of the Colombian weather conditions. We neither incorporate the ENSO phenomenon nor the slight differences of the cyclical pattern of the weather conditions observed each year.

3.4. Model validation

To provide confidence in our simulations and to assess model consistency and robustness, we have exposed our model to standard structural and behavioral tests according to [43,44]. As a result, all validation tests were successfully passed. The results are not shown or discussed here since they are out of the scope of the paper. However, the model robustness is also shown in the advanced sensitivity analysis below, studied in great detail.

In addition, it is noteworthy that we computed the basic time series of our proposed model in Vensim, MATLAB®, and Simulink® to verify their accuracy, and all three of them showed good agreement.

4. Simulation of the Colombian electricity market

The parameter values used for the simulations can be found in Appendix A. The scope and assumptions include:

- The electricity market covers a 33-year time horizon, from 2017 to 2050.
- The market price only considers three different tariffs: prices of variable and permanent generation and a rationing price, based on the average data [45].
- The average hydro and thermal power plants lifetime of 35 and 20 years, respectively. Similarly, an average construction time for hydro and thermal power plants of 5 and 3 years, respectively, is set in the model [35]. It is noteworthy that pipeline or transport delays are used to model the entering and decommission of new power plants, as performed in our previous study [38].
- The start-ups and shut-downs of thermal units are not considered in the model.
- The rationing price is set once the reserve margin drops to zero and maintained if negative values are achieved.
- Thermal plants are modeled with high regulation capacity, while hydro plants are considered without reservoirs and low regulation capacity.
- Market coupling between neighboring countries is neglected since the net imports and exports of electricity in 2017 were less than 1% of the electricity consumption by national users [46].

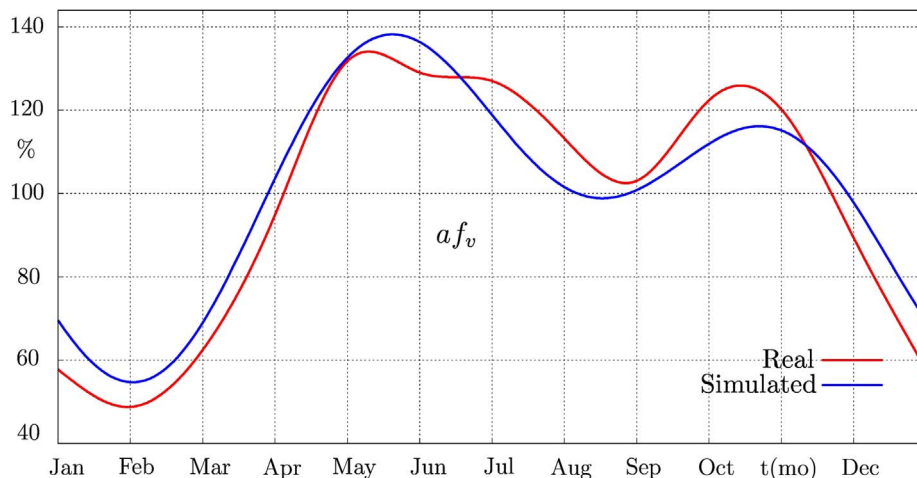


Fig. 6. Availability factor of the variable generation (af_v). The red signal represents the real behavior of the water contributions and is obtained with data from [42]. The blue signal is computed by the authors intended to simulate the real one. The mean square error (MSE) is 1.71%. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

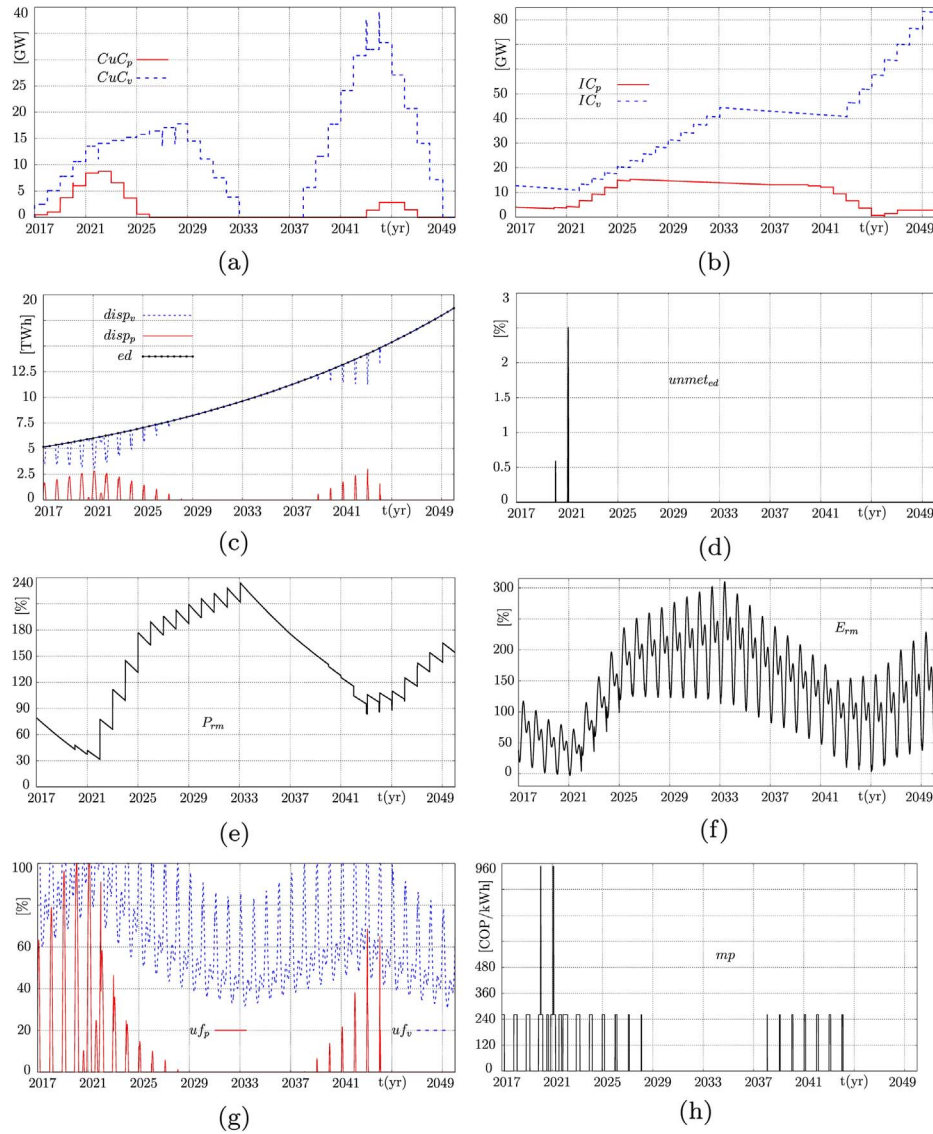


Fig. 7. Simulation of the Colombian electricity market under the BAU scenario. (a) Capacity under construction of (P) and (V) generation (CuC_p and CuC_v , respectively), (b) installed capacity of (P) and (V) generation (IC_p and IC_v , respectively), (c) electricity demand (ed) and dispatch of (P) and (V) generation ($disp_p$ and $disp_v$, respectively), (d) unmet electricity demand ($unmet_{ed}$), (e) power reserve margin (P_{rm}), (f) energy reserve margin (E_{rm}), (g) utilization factor of (P) and (V) generation (uf_p and uf_v , respectively) and (h) market price (mp).

Fig. 7 shows the behavior of the Colombian electricity market under the business-as-usual (BAU) scenario. In general, with the actual policies and the average population growth, hydroelectricity continues to dominate the electricity mix of the Colombian market although the climate variability considerably affects its regulation capacity. Fig. 7(g) shows the utilization factor of the permanent and variable generations (uf_p and uf_v , respectively), which measures the market share of each technology. It is noteworthy that the variability effect of hydroelectricity is introduced in the overall market dynamics; in fact, its participation varies from 35% to 100%, according to the seasonality that is directly transferred to the water contributions (see Fig. 6); consequently, the uf_p is also influenced and exhibits intermittent dynamics, showing periods of low, middle, and high participations to maintain the reliability. Indeed, in 2020 and 2021 it is expected to require the full capacity of the thermal generation (100%) because of the shortfalls of water. It is noteworthy that, although Colombia is primarily a hydro generator country, thermal power is necessary to guarantee the system reliability in case of water deficits. In fact, as shown in Figs. 7(e,f) the reserve margin reaches critical values in these years and also close to 2045; more precisely, the energy reserve margin (E_{rm}) drops to negative

values in the first case and 3.3% in the second one. As a result, it is expected to have unmet electricity demand ($unmet_{ed}$) in 2020 and 2021, as shown in Fig. 7(d). However, although the E_{rm} is neither zero nor negative in 2045 (the system is able to meet the demand and therefore the $unmet_{ed}$ is zero, as shown in Fig. 7(d)), the national electricity market might be under a serious alert during this period. Consequently, the energy system sets a higher market price mp (called the first rationing step) for consumers during the rationing periods (2020 and 2021), and then maintains normal tariffs based on the merit order effect (while E_{rm} remains positive), as shown in Fig. 7(h).

The first problem detected in the Colombian power market is that the mp fails to incentivize the investment in the new capacity in sufficient advance; consequently, the $unmet_{ed}$ is inevitable. The Colombian government should change its investment policies to stimulate signals of investment in the new capacity in a more convenient time.

Furthermore, the dispatch process shown in Fig. 7(c) also reveals how the variability in hydrogeneration necessitates the use of thermal units to fully cover the ed in several situations. However, this is expected to happen when the market is liberalized, and the electricity generation mix is strongly dependent on renewable technologies, either

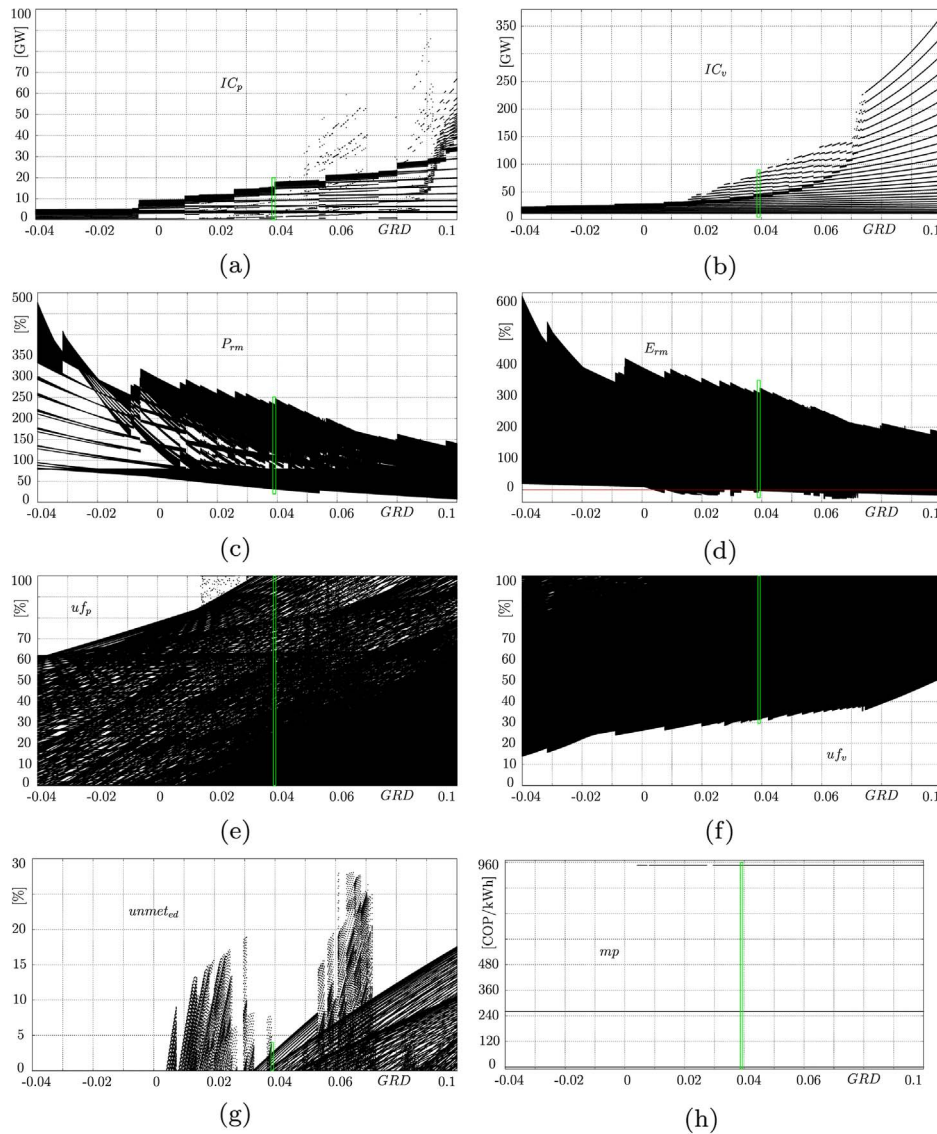


Fig. 8. GRD scenarios. GRD is varied from -0.04 to 0.1 . The green rectangles marked in all figures represent the solution for the Colombian case ($GRD = 0.039$) (a) IC_p , (b) IC_v , (c) P_{rm} , (d) E_{rm} , (e) u_{f_p} , (f) u_{f_v} , (g) $unmet_{ed}$ and (h) mp . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

variable or intermittent. Indeed, although hydrogeneration in Colombia is by far the cheapest generation technology (close to 0 COP/kWh), the thermal power plants drive the market because of their higher mp (300 COP/kWh on average); as a result, the expansion of the variable generation source (see Fig. 7(b)) is larger over time, in accordance with demand growth (see Fig. 7(c)), but not only because of environmental issues, but also owing to high revenues expected. Fig. 7(a) also shows that the new hydro capacity is installed during the first 16 years, while the expansion of thermal capacity lasts 9 years, from 2017 to 2026, according to the real market situation. Nevertheless, if a new variable or permanent capacity is not available to operate before 2020, an electricity crisis will be inevitable during, most likely, 2019, 2020, and 2021 in Colombia. As shown in Fig. 7(b), a significant capacity will enter the system after 2021, while the previous years will only be supplied by the initially installed capacity. As expected, this is not enough to meet the growing demand in a country with 70% of variable generation capacity installed. In other words, Colombia requires significant capacity ready to operate in 2019 by the latest, otherwise the consumers will experience at least two years of energy crisis.

After overcoming the initial 5 years, a safe margin for a long period of time is expected (see Figs. 7(e,f)). In fact, taking into account that the

expansion of the whole energy system in the first 6 years is good enough, there are no incentives for investing in a new capacity until 2038, when Colombia might need additional capacity to avoid a deficit around 2045. Notably, the simulations show that it might be possible to fully cover the ed with only variable generation from 2029 to 2038, as illustrated in Fig. 7(c), or at least a very low participation of thermal power plants can be expected during these years. However, this does not mean that thermal plants will disappear completely, and that Colombia will become a 100% hydro country. On the contrary, one can see that either thermal or other (P) generation sources will be necessary not only in the forties, but also during the complete 33 coming years in order to keep the margin in a safe zone and be able to support the electricity market in case of events of more pronounced variabilities (ENSO phenomenon). In other words, alternatives sources of generation will always be necessary in the Colombian electricity matrix as a support for the hydrogeneration plants and to meet the current reliability conditions.

It is also noteworthy that, as shown by the simulation results, the Colombian market might not be able to reduce significantly its fossil fuel generation parks in the short run, primarily because the variability in hydrogeneration (which becomes much critical during the ENSO

phenomenon) provokes shortfalls that need to be covered by permanent, dispatchable, and trustworthy generation sources. In fact, although the thermal capacity is slowly decreasing (see Fig. 7(b)), some megawatts need to be installed especially in the near future and a few more eventually (see Fig. 7(a)) to prevent electricity rationing. In this sense, the sources of fossil fuel generation are expected to be in the Colombian energy mix for a long period time to come.

In general terms, under the BAU scenario and if a new capacity enters the generation park before 2020 (if *Hidroituango* is not delayed once more, Colombia expects this plant to be ready to operate by the end of 2018 [35]), the Colombian power market is expected to work efficiently without electricity deficits in the long run, especially if we consider a normal *GRD* scenario. The question arises as to what would happen when the electricity demand grows faster or even slower than expected. What would the Colombian electricity market learn when demand grows or decreases unexpectedly? To answer these questions, we shall study the Colombian electricity market under different demand scenarios, based on bifurcation and control theory methods.

5. Demand growth scenarios: a bifurcation perspective

So far, we consider the BAU scenario under normal growth demands to describe the possible dynamics of the Colombian market in the long run. Now let us investigate, using a deeper detailed analysis strategy by virtue of bifurcation theory, not only a low or high growth demand scenario, but also a broader study by simulating the Colombian power market under several growth demand scenarios, with which we ought to be able to cover an extended spectrum of possible demand growth dynamics. It is well known that at the fundamental level, the population, economic growth, and energy efficiency improvements are the key influences on energy demand. In fact, a decrease in the system *ed* is also very likely because of the diffusion of self-generation with solar panels. In this sense, it is important to analyze different *ed* scenarios that can contribute to the Colombian energy planning process. As such, we shall use the parameter *GRD* shown in Fig. 4. This parameter, which represents the population and economic growth, directly influences the demand dynamics. In fact, to perform this analysis we use the Simulink® model shown in Appendix A. Subsequently, we program in a single MATLAB® script a bifurcation algorithm to study the Colombian electricity market dynamics under the *GRD* variations. Accordingly, the *GRD* is varied from -0.04 (-4%) to 0.1 (10%), which is in agreement with the *GRD* changes that Colombia has undergone in the last 27 years [47].

First, let us refer to Fig. 8, where the *GRD* scenarios were computed for the key electricity market variables. In this case, we have obtained a more advanced sensitivity analysis diagrams from the DS perspective, based on the concept of bifurcation diagrams. In DS, bifurcation diagrams are used to determine the stability of a system for different parameter values once the transient behavior has finished. However, electricity market models are characterized for having an infinite transient behavior, implying that they cannot reach stable solutions. This kind of system strongly depends on the growing investment decisions and demand; thus, no stability regimes or clear leverage points can be obtained. Nevertheless, the concept of bifurcation diagrams can be used even for signals with infinite transient responses taking into account all the possible solutions of the variables or system equations; as a result, we can obtain an insight into all the possible output behaviors and their characteristics in one graph. By contrast, with the classic sensitivity analysis diagrams well known in the SD community, our advanced sensitivity analysis provides one remarkable difference, in that the exact parameter value that produces a determined behavior pattern can be obtained. In fact, if someone is interested in the shape of all time series, the classic sensitivity diagrams can be obtained as a complementary source of information. Namely, both methodologies of sensitivity analysis can be used as complementary tools.

According to Fig. 8(a) and (b), clearly, the IC_v is always above IC_p .

As previously explained, by virtue of the merit order effect (which gives hydroelectricity the opportunity to dispatch its energy in the first place) and the high revenues received, more hydro power plants are built in the short and long terms regardless of the *GRD* value. It is noteworthy that for the Colombian case (green rectangle), the IC_p and IC_v reach maximum values of 15 GW and 83 GW, respectively, as is also shown in Fig. 7(b). As shown, IC_v reaches this value almost at the end of the simulation (2049), while IC_p in 2026. In our sensitivity analysis, the confidence limits can be established, but their year of occurrence cannot be inferred from the same graph. However, for negative values of *GRD*, both the IC_p and IC_v do not grow significantly, which is explained by the continuous drop in consumption due to the decreasing *ed*. Moreover, increasing the *GRD* value results in a clear growth in both installed capacities. In particular, the IC_v exhibits an exponential growth while the IC_p presents intermittent ranges with very high amplitudes, such as $GRD \approx (0.05, 0.07)$ and $GRD \approx (0.085, 0.093)$. These sudden increments in the IC_p can be explained by the critical drops of the E_{rm} , as shown in Fig. 8(d). In fact, as shown in Fig. 8(c) and (d), the P_{rm} and E_{rm} remain in a safe zone while the *GRD* is negative. When the *ed* is decreasing, both the P_{rm} and E_{rm} grow higher but will not reach the critical values. However, when we further increase the *GRD* the margin becomes more and more limited. Note that for positive *GRDs*, the E_{rm} drops to negative values, and there are only a few parameter values at which the margin does not cross the red line. For the Colombian case, the E_{rm} crosses the red line exactly in 2020 and 2021 as shown in Fig. 7(d). Nonetheless, the worst-case scenarios could occur around $GRD = 0.02$ and $GRD = 0.065$.

In fact, our electricity market model can undergo $unmet_{ed}$ for several positive *GRD* values as illustrated in Fig. 8(g). For some parameter values, the $unmet_{ed}$ could be very high, while for others it could be really low or even zero. In Colombia, the *GRD* is approximately 0.039, and as shown in Fig. 7(d), the possibilities of $unmet_{ed} \approx 0.65\%$ in 2020 and $unmet_{ed} \approx 2.5\%$ in 2021 exist. This latter value is clearly shown in Fig. 8(g). However, one can see from this picture that the risk of experiencing a larger possible value of $unmet_{ed}$ is imminent even if our *GRD* increases or decreases a little. In other words, Colombia is currently in a safe and risky zone simultaneously, because the $unmet_{ed}$ can become larger for a *GRD* greater than 0.039, and it might also become really high if the *GRD* decreases a little from 0.039. Clearly, the $unmet_{ed}$ can be avoided if we obtain a *GRD* close to 0.0286 and 0.0078. Nevertheless, the possibility to reduce the *GRD* in the short term is slight, since it can only be achieved by incentivizing the self-generation using solar panels from either residential or industrial consumers and increasing the energy efficiency. Even though Colombia approved Law 1715 in 2014 for promoting the development and use of non-conventional renewable technologies, their deployment is still very slow owing to economic and knowledge issues.

On the contrary, Colombia is expected to have a larger *GRD* over time, which means that every year this country could be subjected to an increase in the risk of undergoing significant $unmet_{ed}$. Hence, this country can only avoid such risks by increasing the installed capacity of its electricity mix before 2019; otherwise, the country might experience a serious energy problem in the short term. Moreover, from Fig. 8(h) one can see that the $unmet_{ed}$ can also be detected in the *mp*. Recall that when the E_{rm} becomes zero or negative, the system sends a signal of rationing price; therefore, one can see in the *mp* that the signal of rationing price appears in more than a 60% of the total parameter values. In particular, as it is also shown in Fig. 8(c), (d), and (g), under negative *GRD* values, our model exhibits good conditions to operate and meets the *ed* without restrictions; namely, the *mp* only takes the values between the price set by the thermal generation and the price set by the hydrogeneration. On the positive side, only three scenarios result in the avoidance of the rationing price, i.e., close to $GRD = 0.0016$, $GRD = 0.0078$, and $GRD = 0.0286$, which are exactly the regions shown also in the $unmet_{ed}$ (see Fig. 8(g)); in the other regions the rationing signal is sent at some time between 2017 and 2050.

Additionally, Figs. 8(e,f) illustrate the participation of each generation technology. As shown, on one hand the hydro power plants (u_v) are utilized in all scenarios ranging from not less than 14% and up to 100%. However, this interval of participation shrinks for larger GRD values. For instance, in the Colombian case the u_v ranges from 31% to 100% depending on different conditions, while for a $GRD = 0.1$ one would expect the u_v to range from 51% to 100%, which is 20% less participation than the Colombian case. Evidently, larger GRD values involve more electricity to meet the increased demand; consequently, the participation of the hydro power plants must start with higher values. On the other hand, thermal power plants are not used in some events (where hydro generation meets the required ed by itself) and therefore provide a $u_p = 0\%$, as shown for any GRD value. More precisely, in the interval $GRD = (-0.04, 0.014)$, thermal power plants are used for no more than 61% and 85%, respectively, while in the interval $GRD = (0.014, 0.1)$, the u_p ranges from 0% to 100%. This means that when necessary, the full potential of the thermal plants is dispatched to support the hydrogeneration, meet the demand, and avoid rationing events; however, this is sometimes inevitable as shown in Fig. 8(e) and (g).

6. Confidence limits and their occurrence: a bifurcation perspective

So far, we have considered the advanced sensitivity analysis from the DS perspective. This analysis is very important to explore the possible scenarios of the Colombian electricity market in case of changes in the GRD . Although we can have a deeper insight into the primary variables under these changes, we also found that it is necessary to determine the exact year of occurrence of the confidence limits and thus provide a more complete information of the scenarios. Once again, this can be done by implementing a detection algorithm in a single MATLAB® script. It is noteworthy that this kind of algorithm cannot be implemented in the common SD software packages.

As shown in Fig. 9, under the DS perspective more information can be obtained from the SD model. In this case, we illustrate the confidence limits along with their corresponding year of occurrence. In other words, for every variable its possible maximum and minimum values were plotted together with the exact time information, for each GRD scenario.

In principle, Fig. 9(a) shows that the IC_p in the GRD range $(-0.04, 0.05)$, including the Colombian case, reaches a maximum value approximately in 2024, and after that it loses its capacity until it reaches a minimum value approximately in 2044. This fact suggests that if the GRD takes on values within the range mentioned previously, the thermal capacity might tend to increase its power in the first 6 years. Hereinafter, no more incentives are expected to expand the thermal parks; as a result, their capacity might be subjected to drop for at least 20 years. As explained in the BAU scenario of the Colombian case, the margin is expected to drop approximately in 2043 and grow again there on (see Fig. 7(e) and (f)). For this reason, the IC_p exhibits its lowest value almost at the end of the simulation, as shown in Fig. 7(b) for the Colombian case. We can expect this same behavior pattern for all GRD values within $(-0.04, 0.05)$. However, for GRD values in the range $(0.05, 0.07)$, although the minimum thermal capacity is also reached around 2043, the maximum thermal capacity can occur close to 2049 owing to the appearance of the large $unmet_{ed}$ events that would occur at this time (see Fig. 9(g)), causing an increase in the number of investment decisions for more thermal capacity. Further, in the GRD range $(0.07, 0.086)$ the events of $unmet_{ed}$ are once again not expected at the end of the simulation; therefore, the behavior pattern explained for the range $(-0.04, 0.05)$ applies here as well. Finally, if we consider the GRD values within the range $(0.086, 0.1)$, a different behavior pattern is observed. The minimum thermal capacity is now detected close to 2020 while the maximum capacity appears around 2049. In this case, unlike the GRD interval $(0.05, 0.07)$, where $unmet_{ed}$ events cause a significant

increase in the IC_p at the end of the simulation, the maximum capacity here is achieved owing to the larger demand scenarios, which cause the IC_p to grow faster; consequently, the minimum capacity is detected here at the critical events expected to occur around 2020, at which situations of $unmet_{ed}$ are also presumed.

On the contrary, the IC_v shown in Fig. 9(b) illustrates that its capacity will not drop significantly in the long run. In fact, one can see that regardless of the GRD scenario, the minimum IC_v is always expected to occur around 2022. This is due to the depreciation of the initial installed capacity, which subtly affects its value during the first 4 or 5 years, and is expected to finish around 2022 (where the minimum IC_v is detected) once a new variable power is poised to enter the market. Additionally, for GRD values within $(-0.4, 0.015)$, the maximum IC_v is achieved close to 2029. This behavior pattern is attributed to the P_m or the E_{rm} , which do not grow significantly in the long run owing to the slow growth demand scenarios. In fact, for the negative GRD values, both the P_m and E_{rm} tend to remain in high levels by virtue of a decreasing demand. In this sense, the maximum IC_v is achieved in the short or middle terms owing to the new capacity built to overcome the critical events, which are presumed to occur in the twenties, approximately 5 years before the IC_v achieves its maximum capacity. Conversely, for GRD values within the interval $(0.015, 0.1)$, the maximum IC_v is typically achieved at the end of the simulation, close to 2049. For these scenarios the demand is large enough to provoke critical values of the E_{rm} in prior years; as a result, the necessary investments in a new capacity are performed earlier. Subsequently, the IC_v reaches its maximum value 5 years after (around 2049).

Furthermore, as shown in Fig. 9(c) and (d), the P_m and E_{rm} differ by a number of properties. In general terms, the P_m always exhibits its lowest values in the short run, around 2020, because of the depreciation of the power plants, which is expected to significantly affect the margin until the first new capacity comes into operation. Additionally, our electricity market model is expected to attain the maximum P_m , principally, in the middle term. This means that all power plants that will be built to support the critical situations expected to occur in the twenties, will provide the maximum P_m a few years later. Conversely, the E_{rm} in Colombia is highly affected by weather conditions; consequently, its confidence limits occur at different points in time. The minimum E_{rm} is detected three intervals in the short run. In the first interval, predominates negative values of GRD and $unmet_{ed}$ are avoided although the E_{rm} maintains its critical values. Further, the maximum E_{rm} is achieved, as expected, in the middle run owing to the investments carried out in the critical twenties, but it approaches to the last years as the GRD gets smaller (a larger decrease in demand gives rise to continuously increasing margins). Unlike the first interval, in the second one ($GRD \approx (0.03, 0.055)$), the $unmet_{ed}$ cannot be prevented and rationing events are also expected to occur in the short run (see also Fig. 9(g)). Similar to the previous case, here the maximum E_{rm} is detected in the middle run, but it also approaches the long run as GRD increases, owing to the significant investments in the new capacity carried on during the critical twenties. However, in the third interval ($GRD \approx (0.055, 0.1)$), the maximum values of E_{rm} are detected earlier close to 2026 and remain located in almost the same year for every GRD scenario. This behavior pattern depends, in part, on the investments in the new capacity performed during the critical twenties, and also because the demand scenarios grow faster causing a rapid drop in the margin during the middle and long runs. Finally, if one looks at the other intervals, by contrast, the minimum values of the E_{rm} occur at the last years, including the Colombian case, and the maximum values share the same characteristics of the already explained previous cases.

In addition, Fig. 9(e) and (f) illustrate the confidence limits of the u_p and u_v , respectively. As shown clearly, the u_p shows that thermal generation would not be entirely used for negative and small values of the GRD . In fact, although one could expect that thermal generation would not be necessary for small or negative values of the GRD , simulations show that thermal power is crucial in all GRD scenarios to

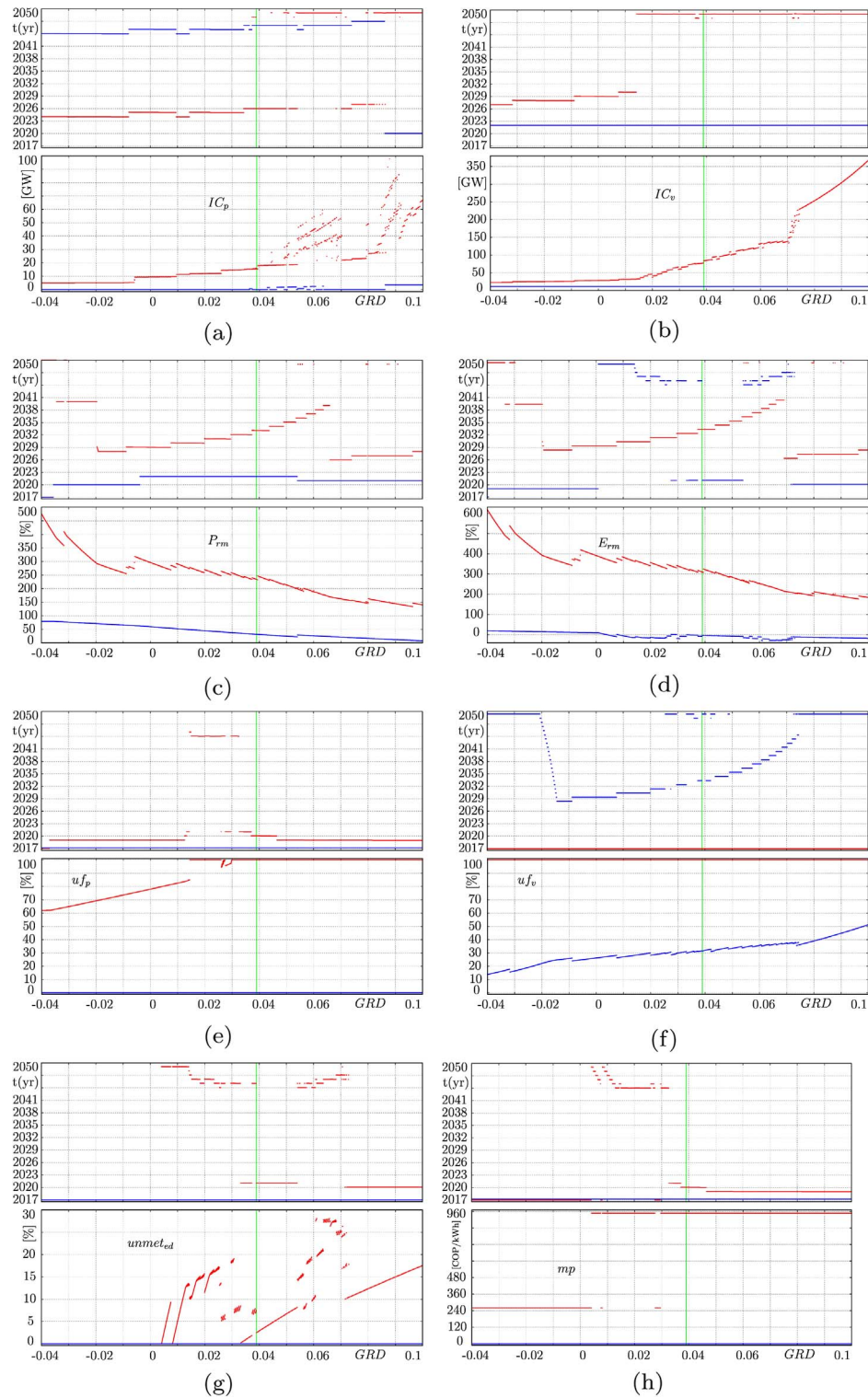


Fig. 9. Confidence limits of the GRD scenarios together with the time of occurrence. The marked green vertical line represents the Colombian case ($GRD = 0.039$) (a) IC_p , (b) IC_v , (c) P_m , (d) E_r , (e) u_f , (f) u_v , (g) $unmet_{ed}$ and (h) mp . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

support the variability in hydrogeneration, especially in the short run where it is very likely to have a significant electricity risk. Indeed, the u_f shows that the participation of thermal plants oscillates between 0% and 100%. In particular, their full potential will be used in the majority of the positive GRD values in the short run, approximately in 2019. This suggests again that 2019, 2020 or 2021 can be very likely considered the critical years if a significant new capacity does not enter to operate in the Colombian electricity market before 2019, regardless of the GRD

value. Only if the GRD value decreases to within the interval $GRD \approx (0.013, 0.03)$, will a critical situation in the long run be likely, around 2044. On the contrary, the u_v shows that hydro power plants are always used in the first place to cover the ed . In the worst case, their participation is only 15–22% around 2049 if the GRD takes on values within the range $(-0.04, -0.02)$. The same behavior pattern is expected to occur for GRD values within $(0.072, 0.1)$, with participations of 37–51%. It is noteworthy that the minimum participation that the u_v

can achieve for the other GRD values practically occurs in the middle and long runs, except for some GRD values within the range (0.025, 0.05), where the minimum values can be achieved around 2049. Moreover, a full participation of hydro plants is expected from the very beginning of the simulation and then ranging from approximately 30% to 100%.

Finally, Fig. 9(g) and (h) show the $unmet_{ed}$ and mp , respectively. As shown, for negative values of the GRD and a short range of positive ones, events of electricity rationing can be discarded. Consequently, taking into account that both thermal and hydro plants are dispatching electricity to meet the demand, the mp that consumers perceive are mostly set by thermal producers. Subsequently, possible electricity rationing are expected to occur in the rest of the GRD values, except for GRD values close to 0.007 and 0.028, where the $unmet_{ed}$ can also be avoided and the mp is also mostly set by thermal producers. In general, the most serious cases can occur for GRD values in the ranges (0.0332, 0.03732), (0.03896, 0.054), and (0.07168, 0.1), where electricity rationing is expected in the short run, around 2020. Consequently, consumers experience an mp set by the rationing price during some periods in 2020. On the contrary, the other GRD values avoid serious problems in the short run. Instead, the worst cases are likely to occur in the long run, around 2044.

7. Detailed rationing events: a control theory perspective

Now let us consider in more detail, all the possible rationing events of the GRD scenarios. As discussed and shown above, some GRD scenarios might undergo $unmet_{ed}$ or rationing events. The question arises as to whether it is possible to estimate the number of months expected to be under electricity rationing, their specific year or even the exact month.

Based on the DF concept, which is a typical method to design controllers for nonlinear systems [48], we have developed a graphical tool that can specifically determine how many months are expected to undergo rationing events, their year and month of occurrence, and even the probability of such episodes. To develop this graphical tool, we also use a single MATLAB® script to program the corresponding algorithm.

Fig. 10 shows the DF diagram computed for every GRD scenario. As shown, the bottom left diagram illustrates the number of rationing months (FRM) that might exhibit each GRD scenario. In fact, their corresponding year and month of occurrence can be estimated with the

upper left and right diagrams. Furthermore, with the bottom right diagram one can determine the probability of occurrence of every rationing episode, i.e., the exact month of rationing.

In particular, we observed that Colombia might exhibit two months of electricity rationing (see green horizontal line). If we look at the mapped upper diagram, it shows that these two months might take place in 2020, 2021, 2044, 2045, 2048 or 2049. However, we can verify from the BAU Colombian scenario shown in Fig. 7(d), that events of $unmet_{ed}$ are expected to occur in 2020 and 2021. Accordingly, we can certainly conclude that Colombia might undergo one month of electricity rationing in 2020 and one more in 2021. Subsequently, from the corresponding mapped upper right diagram, one can see that these two months might refer to January, February, November or December. More precisely, we can see again in Fig. 7(d) and observe very carefully that the rationing events take place at the very beginning of the years; as a result, we can affirm that the two months of electricity rationing expected to occur in 2020 and 2021 in Colombia are very likely to be both in January. Finally, if we look at the bottom right of the diagram, one can see the corresponding 3D diagram. The latter shows that January is in fact a very dangerous month because its probability of occurrence is close to 25% in 2020 and 30% in 2021. Indeed, these two rationing events show the highest probability of occurrence, which means that Colombia must consider seriously the years and months mentioned here to avoid such critical situations very likely to occur in the short run. Once again, our model shows that Colombia clearly needs new power to be installed before 2020 if the serious rationing events mentioned above are to be avoided.

Nevertheless, the negative GRD scenarios do not show the FRM , whereas the positive values exhibit the FRM ranging from 1 to 12 months. As the GRD value becomes larger, the FRM increases as well. Nevertheless, it is noteworthy that, in general, regardless of the GRD scenario, the rationing events are clearly expected to happen in the short and long runs, around 2020 and 2045. In fact, we observed that January, February, and December tend to be the more critical months. In particular, December and January are likely to be the most severe months of the rationing events with probabilities of occurrence ranging from 10% to 30%, respectively, although February, March, and November might also be critical for other GRD scenarios.

Our explanation above also lets us deduce that the critical situations expected to occur in the short run basically determine the safe time period (from 2022 to 2041) of the Colombian electricity market in the

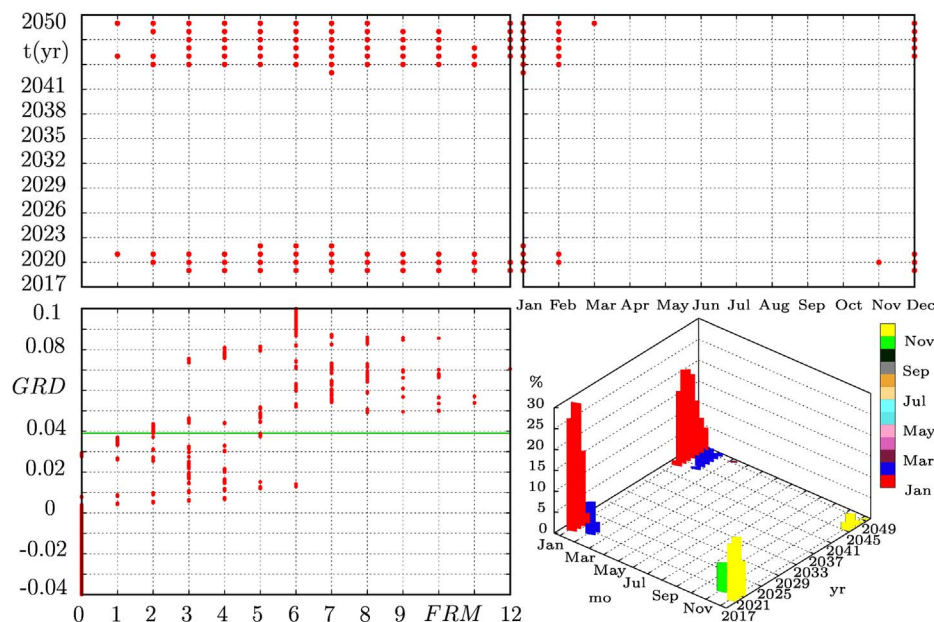


Fig. 10. DF diagram of the GRD scenarios. FRM stands for frequency of rationing months.

middle run. In other words, the rationing events expected to occur in the short coming years incentivize investments in new capacity to overcome such situations. Although it might be late for the short run cases, it is expected that these investments also provide enough power to the Colombian electricity market to guarantee good market conditions in the middle run, as shown in the upper left diagram of Fig. 10. Moreover, as shown in Fig. 6, January, February, March, November, and December are characterized by the sunniest months in Colombia owing to seasonality issues; accordingly, these months are also reflected as the critical events of all *GRD* scenarios, as shown in the upper right diagram of Fig. 10. This clearly suggests that seasonality strongly affects the Colombian electricity market, with a defined behavior pattern, regardless of its *GRD* scenario.

8. Conclusions

Colombia might be subjected to different *GRD* scenarios in the short and long runs. In fact, it is well known that the growth or decrease in demand may primarily depend on the GDP (gross domestic product) changes, economic recession, population growth, energy efficiency, microgeneration or electricity tariff. All these factors are rapidly changing over time in Colombia, and one of them, or several of them might bring down or up the *ed* of the country. As a result, different *GRD* scenarios are evaluated in this paper to provide a deeper insight and a detailed description of the possible realities that Colombia could face in the near or long future.

In fact, the present paper describes, in great detail, the dynamics of several key variables of the Colombian electricity market under a broad spectrum of *GRD* scenarios. Specifically, the overall market performance is investigated by combining the SD and DS approaches, with which it was able to carefully address the nonlinearities and feedback relations of the system variables. In particular, our paper shows that Colombia is under a serious risk in the near future. If significant new capacity does not enter the power market before 2019, Colombia might be subjected to rationing events in 2020 and 2021. More precisely, our analysis suggests that January, February, November and December are most likely the most susceptible months to undergo *unmet_{ed}*. Indeed, if Colombia maintains its current *GRD* scenario, it is very likely that January becomes the most dangerous month in both 2020 and 2021. However, this paper also reveals that regardless of the *GRD* scenario, the variability in Colombia's weather causes the rationing events to occur, initially, close to 2020, and eventually, close to 2045. Although Colombia is preparing for a new large hydro capacity to enter (initially thought for 2017) in the late 2018 [35], our results recommend no more delays in the construction of this hydro plant if rationing events are to be avoided.

Nevertheless, our results indicate that the variability in hydro-generation not only affects its dispatching process, but also the generation of the permanent technology. In fact, it was observed that this variability is indirectly transferred to all the market variables, although with a lower level in some of them. Consequently, uncertainty in meeting the demand can be expected. More importantly, other sources

of generation need to be in the electricity matrix to support the variable hydro and mitigate its direct consequences. However, this might not be sufficient as shown in this paper. Suitable investment decisions must be carried out. Although we only considered a simplified version of the Colombian electricity market and a perfect cyclical pattern (an approximation to the real Colombian seasonality) of its weather, our model successfully captures its main properties and dynamics, together with several counter-intuitive behaviors. Nevertheless, the ENSO phenomenon will be included in the a_f in future research to make our model more realistic. It is also noteworthy that some of the lessons learned here can be applied to other energy markets of other countries.

Finally, by combining SD and DS, we are able to develop more advanced analysis tools that lead us to obtain more complete results, with which deeper insights into the systems dynamics complexity can be attained. In particular, based on the bifurcation theory, a more sophisticated sensitivity analysis was developed. Remarkably, one of them also lets us determine the confidence limits and their year of occurrence. Moreover, inspired by the DF, which is typically used to design controllers for nonlinear systems, we also developed a graph that allows us to determine, with great detail, the rationing events undergone by a broad spectrum of the *GRD* scenarios. This graph even provides information about rationing events in terms of months, together with the probability of occurrence. According to this, we believe our paper will impact significantly the way SD modelers and energy specialists study or investigate the current problems that do not allow the appropriate development of renewable energies. Under our proposed methodology, better policies and market structures will be formulated since a broader/deeper analysis can be performed.

It is also noteworthy that transforming the stock–flow structure in a Simulink block diagram has shown to be the easiest way to access any DS tool, or for going even further than that. This transformation is very simple (as shown in [15]) and no mathematical equations are needed. The same steps applied in Vensim, Stella or Powersim to formulate the stock–flow structure can be applied in Simulink, however, some functions or definitions can change because it is being used in a different software package. In fact, regardless of the abstraction level of the SD models, or how big they are, Simulink gives even more advantages than the common SD softwares. However, we want to emphasize that it is mandatory to construct the stock–flow structure in any SD software as usual, before resorting to Simulink.

Finally, this work studies very deeply an electricity market model using novel tools and strategic insights never reported in the literature, which provides further understanding of its complex behaviors. Therefore, we encourage the SD modelers, academics, and practitioners interested in long-term planning models to implement and adopt our proposed analysis methodology.

Acknowledgments

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Appendix A

Table 1

Table 1
Parameters of the Colombian electricity sector.

Parameter	Value
(V) Construction time (CT_v)	5 yr
(P) Construction time (CT_p)	3 yr
(V) Lifetime (LT_v)	30 yr
(P) Lifetime (LT_p)	20 yr
Growth rate of demand (GRD)	0.039
(V) Variable cost (VC_v)	14.9829 COP/kWh
(P) Variable cost (VC_p)	47.543 COP/kWh
(V) Incentives (I_v)	0 COP/kWh
(P) Incentives (I_p)	0 COP/kWh
(V) Variability fixed cost (VCV)	41.3516 COP/kWh
(P) Variability fixed cost (VCp)	63.7944 COP/kWh
(V) Capacity under construction(0)	0 MW
(P) Capacity under construction(0)	0 MW
(V) Installed capacity(0)	11611.1 MW
(P) Installed capacity(0)	4833 MW
Power demand(0)	9320 MW
(V) price	0.001 COP/kWh
(P) price	261 COP/kWh
Rationing price	947 COP/kWh
(P) Availability factor	92.5%
Capacity factor	70%
Elasticity of demand (ϵ)	-0.03

A.1. Simulink block diagrams

Figs. 11–13.

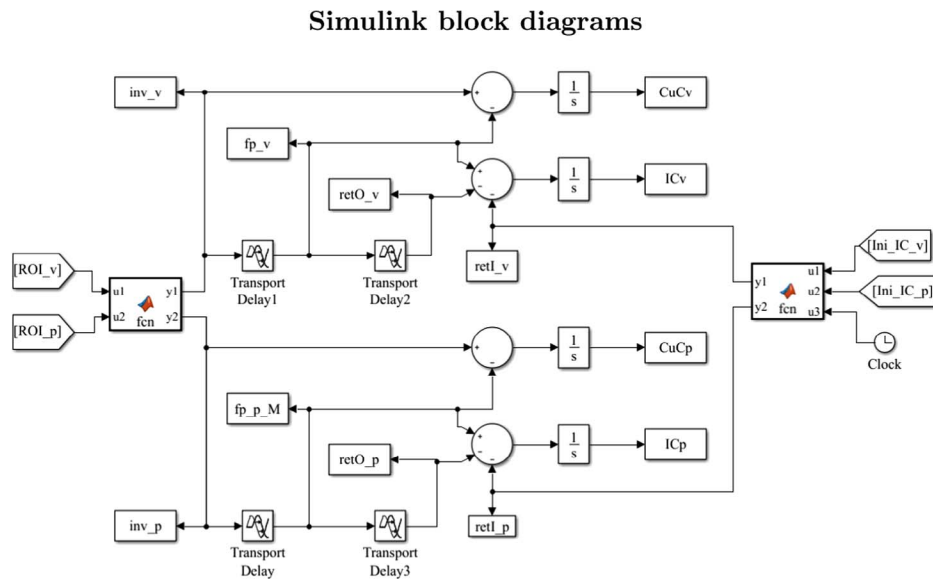


Fig. 11. Supply side from (P) and (V) generation.

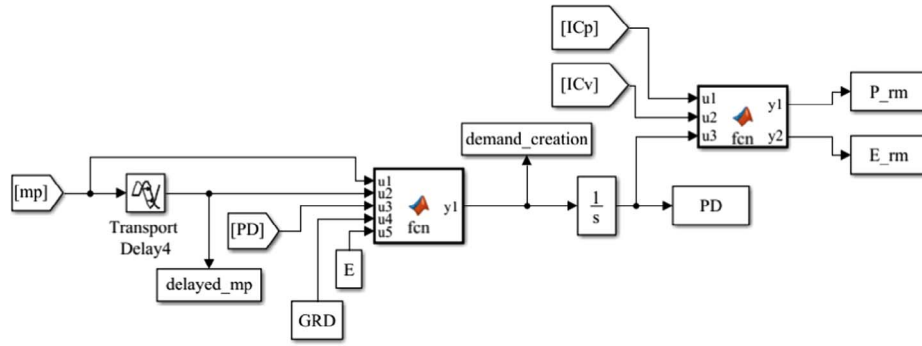


Fig. 12. Demand component.

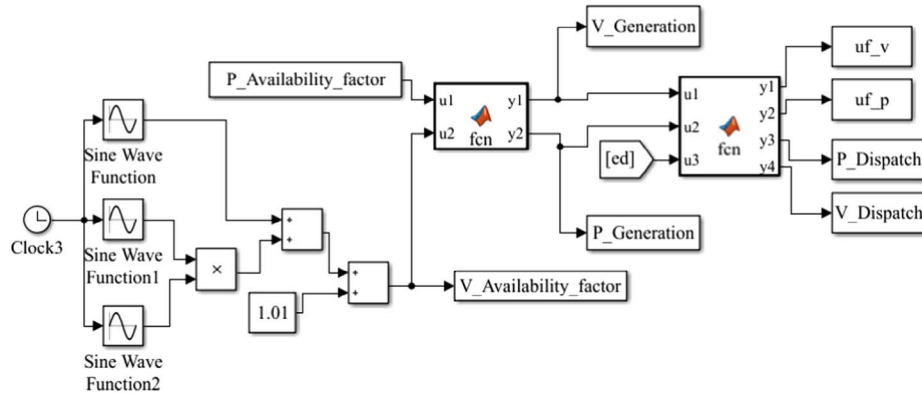


Fig. 13. Electricity dispatch.

A.2. System equations

$$\begin{aligned}
 CuC_p &= CuC_p(0) + \int_0^t (inv_p - fp_p) \cdot dt \\
 IC_p &= IC_p(0) + \int_0^t (fp_p - rop_p - rip_p) \cdot dt \\
 CuC_v &= CuC_v(0) + \int_0^t (inv_v - fp_v) \cdot dt \\
 IC_v &= IC_v(0) + \int_0^t (fp_v - rop_v - rip_v) \cdot dt \\
 PD &= PD(0) + \int_0^t dc \cdot dt
 \end{aligned} \tag{2}$$

$$inv_p = \begin{cases} 0 & ROI_p \leq 0 \\ \frac{PD}{PD(0)} k_1 \delta(t) & 0 < ROI_p \leq 10 \\ \frac{PD}{PD(0)} k_2 \delta(t) & ROI_p > 10 \end{cases}$$

$$inv_v = \begin{cases} 0 & ROI_v \leq 0 \\ \frac{PD}{PD(0)} k_1 \delta(t) & 0 < ROI_v \leq 10 \\ \frac{PD}{PD(0)} k_2 \delta(t) & ROI_v > 10 \end{cases} \tag{3}$$

$$\begin{aligned}
 fp_p &= inv_p(t - CT_p) \\
 fp_v &= inv_v(t - CT_v) \\
 rop_p &= fp_p(t - LT_p) \\
 rop_v &= fp_v(t - LT_v)
 \end{aligned} \tag{4}$$

$$\begin{aligned}
 rip_p &= \begin{cases} IC_p(0)/LT_p; & t \leq 2017 + LT_p \\ 0 & \text{other case} \end{cases} \\
 rip_v &= \begin{cases} IC_v(0)/LT_v; & t \leq 2017 + LT_v \\ 0 & \text{other case} \end{cases}
 \end{aligned} \tag{5}$$

$$ROI_p(t) = \frac{(mp \cdot uf_p - VC_p - VFC_p + I_p)}{VFC_p + VC_p} 100\%$$

$$ROI_v(t) = \frac{(mp \cdot uf_v - VC_v - VFC_v + I_v)}{VFC_v + VC_v} 100\% \quad (6)$$

$$dc = GRD \times epd \times PD$$

$$epd = \begin{cases} 1; & dmp = 0 \\ (mp/dmp)^{\epsilon}; & \text{other case} \end{cases}$$

$$dmp = mp(t-0.25)$$

$$mp = \begin{cases} P_v; & (gen_v \geq ed \wedge rm > 0) \vee (gen_v < ed \wedge gen_p \leq 0 \wedge rm > 0) \\ P_p; & gen_v < ed \wedge rm > 0 \wedge gen_p > 0 \\ RAP; & (gen_v \geq ed \wedge rm \leq 0) \vee (gen_v < ed \wedge rm \leq 0) \end{cases} \quad (7)$$

$$uf_p = \begin{cases} 0; & gen_p = 0 \\ disp_p/gen_p; & \text{other case} \end{cases}; \quad uf_v = \begin{cases} 0; & gen_v = 0 \\ disp_v/gen_v; & \text{other case} \end{cases} \quad (8)$$

$$disp_p = \begin{cases} 0; & gen_v \geq ed \\ ed - gen_v; & gen_p + gen_v \geq ed; \\ gen_p; & \text{other case} \end{cases}; \quad disp_v = \begin{cases} ed; & gen_v \geq ed \\ gen_v; & gen_v < ed \end{cases} \quad (9)$$

$$P_{rm} = (((IC_p + IC_v) - PD)/PD) 100\%; \quad E_{rm} = (((gen_p + gen_v) - ed)/ed) 100\% \quad (10)$$

$$gen_p = IC_p \cdot Af_p \cdot 30 \cdot 24; \quad gen_v = IC_v \cdot af_v \cdot 30 \cdot 24; \quad ed = PD \cdot 0.7685 \cdot 30 \cdot 24 \quad (11)$$

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